

Title Goes Here:

A VAR-based Computational Analysis of Influenza and Weather
Dynamics

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Abstract

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A Supplementary Methods: Mathematical Details

In this appendix we have the mathematical details of the methods used: vector autoregression (VAR), impulse response functions (IRFs), Granger causality tests, and residual bootstrap procedures.

A.1 Vector Autoregression (VAR)

A.1.1 Model specification

Let $\{\mathbf{y}_t\}_{t=1}^T$ be a K -dimensional time series, where $\mathbf{y}_t \in \mathbb{R}^K$ are all the variables (e.g., influenza counts and weather covariates) at time t . A VAR(p) model with intercept can be written as

$$\mathbf{y}_t = \mathbf{c} + \sum_{i=1}^p A_i \mathbf{y}_{t-i} + \mathbf{u}_t, \quad t = 1, \dots, T, \quad (1)$$

where

- $\mathbf{c} \in \mathbb{R}^K$ is a vector of intercepts (can be modified to add deterministic constants)
- $A_i \in \mathbb{R}^{K \times K}$ are autoregressive coefficient matrices,
- $\mathbf{u}_t \in \mathbb{R}^K$ is the innovation (error) term.

We assume

$$\mathbb{E}[\mathbf{u}_t] = \mathbf{0}, \quad \mathbb{E}[\mathbf{u}_t \mathbf{u}_s^\top] = \begin{cases} \Sigma_u, & t = s, \\ 0, & t \neq s, \end{cases} \quad (2)$$

so that $\{\mathbf{u}_t\}$ is simply random noise.

A.1.2 Deterministic regressors

We can write eq. (1) with deterministic regressors. Let $\mathbf{d}_t \in \mathbb{R}^M$ be a vector of deterministic terms (e.g., constant, linear trend, seasonal dummies, sine/cosine pairs), and let $B \in \mathbb{R}^{K \times M}$ be its coefficient matrix. Then

$$\mathbf{y}_t = B \mathbf{d}_t + \sum_{i=1}^p A_i \mathbf{y}_{t-i} + \mathbf{u}_t. \quad (3)$$

The pure-intercept case corresponds to $\mathbf{d}_t \equiv \mathbf{1}$ and $M = 1$.

A.1.3 Stacked regression form

For estimation, we rewrite the VAR as a multivariate linear regression.

Define the *effective sample size* $T^* := T - p$ since the first few lags cannot be used. Now we can put all the dependent variables from $t = p + 1$ to T into a matrix

$$Y = \begin{bmatrix} \mathbf{y}_{p+1}^\top \\ \mathbf{y}_{p+2}^\top \\ \vdots \\ \mathbf{y}_T^\top \end{bmatrix} \in \mathbb{R}^{T^* \times K}. \quad (4)$$

Next, define the regressor matrix

$$X = \begin{bmatrix} \mathbf{x}_{p+1}^\top \\ \mathbf{x}_{p+2}^\top \\ \vdots \\ \mathbf{x}_T^\top \end{bmatrix} \in \mathbb{R}^{T^* \times (Kp+M)}, \quad (5)$$

where each row is

$$\mathbf{x}_t^\top = [\mathbf{y}_{t-1}^\top, \mathbf{y}_{t-2}^\top, \dots, \mathbf{y}_{t-p}^\top, \mathbf{d}_t^\top] \in \mathbb{R}^{1 \times (Kp+M)}. \quad (6)$$

We also stack the coefficient matrices into a single matrix

$$B = \begin{bmatrix} A_1^\top \\ A_2^\top \\ \vdots \\ A_p^\top \\ B^\top \end{bmatrix} \in \mathbb{R}^{(Kp+M) \times K}. \quad (7)$$

The innovations are stacked as

$$U = \begin{bmatrix} \mathbf{u}_{p+1}^\top \\ \mathbf{u}_{p+2}^\top \\ \vdots \\ \mathbf{u}_T^\top \end{bmatrix} \in \mathbb{R}^{T^* \times K}. \quad (8)$$

Then eq. (3) for $t = p + 1, \dots, T$ becomes the matrix equation

$$Y = XB + U. \quad (9)$$

A.2 Estimation via Ordinary Least Squares

The ordinary least squares (OLS) estimator of B minimizes $\|Y - XB\|_F^2$, the sum of squared residuals over all equations:

$$\hat{B} = \arg \min_B \|Y - XB\|_F^2. \quad (10)$$

Assuming X has full column rank, the solution is

$$\hat{B} = (X^\top X)^{-1} X^\top Y. \quad (11)$$

The fitted values and residuals are

$$\hat{Y} = X\hat{B}, \quad \hat{U} = Y - \hat{Y}. \quad (12)$$

Let $\hat{\mathbf{u}}_t$ denote the row of $\hat{U}$ corresponding to time t .

An estimator of the innovation covariance matrix Σ_u is

$$\hat{\Sigma}_u = \frac{1}{T^* - (Kp + M)} \hat{U}^\top \hat{U} \in \mathbb{R}^{K \times K}, \quad (13)$$

where the denominator uses the residual degrees of freedom. Here M denotes the number of deterministic regressors included in the VAR (e.g., constant, trend, seasonal dummies, or other non-stochastic terms).

Rank-deficient regressors and the Moore–Penrose pseudoinverse

If X does *not* have full column rank, then the matrix $(X^\top X)$ is singular and the closed-form expression (11) is not defined. In this case the OLS estimator is solved using the Moore–Penrose pseudoinverse X^+ :

$$\hat{B} = X^+ Y, \quad (14)$$

which gives us the minimum-norm solution to the least-squares problem $\min_B \|Y - XB\|_F^2$ even if X is rank-deficient. The pseudoinverse is computed via singular value decomposition (SVD).

A.3 Moving-Average Representation and IRFs

A.3.1 Companion form

Let Kp -dimensional stacked state vector

$$\mathbf{Y}_t = \begin{bmatrix} \mathbf{y}_t^\top & \mathbf{y}_{t-1}^\top & \cdots & \mathbf{y}_{t-p+1}^\top \end{bmatrix}^\top \in \mathbb{R}^{Kp}. \quad (15)$$

The VAR(p) can be written in first-order form as

$$\mathbf{Y}_t = \mathbf{C} + F\mathbf{Y}_{t-1} + \mathbf{U}_t, \quad (16)$$

where the *companion matrix* $F \in \mathbb{R}^{Kp \times Kp}$ is

$$F = \begin{bmatrix} A_1 & A_2 & \dots & A_{p-1} & A_p \\ I_K & 0 & \dots & 0 & 0 \\ 0 & I_K & \dots & 0 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & \dots & I_K & 0 \end{bmatrix}, \quad (17)$$

$\mathbf{C} \in \mathbb{R}^{Kp}$ contains the intercepts (and deterministic terms), and

$$\mathbf{U}_t = \begin{bmatrix} \mathbf{u}_t^\top & \mathbf{0}^\top & \dots & \mathbf{0}^\top \end{bmatrix}^\top \in \mathbb{R}^{Kp}. \quad (18)$$

If the VAR is stable, then $(I_{Kp} - F)$ is invertible, and the mean is

$$\mathbb{E}[\mathbf{Y}_t] = (I_{Kp} - F)^{-1} \mathbf{C} =: \boldsymbol{\mu}_Y. \quad (19)$$

$\boldsymbol{\mu}_Y$.

A.3.2 Infinite-order moving-average representation

Under stability, we can rewrite the VAR in *vector moving-average* (VMA(∞)) form:

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \Psi_h \mathbf{u}_{t-h}, \quad (20)$$

where $\boldsymbol{\mu} \in \mathbb{R}^K$ is the unconditional mean of $\mathbf{y}_t$ and $\Psi_h \in \mathbb{R}^{K \times K}$ are the VMA coefficient matrices. We set

$$\Psi_0 = I_K, \quad (21)$$

and find Ψ_h recursively from the VAR coefficients:

$$\Psi_h = A_1 \Psi_{h-1} + A_2 \Psi_{h-2} + \dots + A_p \Psi_{h-p}, \quad h \geq 1, \quad (22)$$

where $\Psi_h = 0$ for $h < 0$.

A.3.3 Structural shocks and identification

The reduced-form innovations $\mathbf{u}_t$ are usually correlated:

$$\mathbb{E}[\mathbf{u}_t \mathbf{u}_t^\top] = \Sigma_u, \quad \Sigma_u \text{ symmetric positive definite.} \quad (23)$$

To interpret shocks as orthogonal or “structural” shocks, we factor Σ_u as

$$\Sigma_u = P P^\top, \quad (24)$$

where P is a nonsingular $K \times K$ matrix. We can do this using the Cholesky factorization with P lower triangular, which encodes some sort of ordering of the variables. We can then write

$$\mathbf{u}_t = P\boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(\mathbf{0}, I_K), \quad (25)$$

so that $\boldsymbol{\varepsilon}_t$ has uncorrelated unit-variance. Substituting into eq. (20) gives

$$\mathbf{y}_t = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \Psi_h P \boldsymbol{\varepsilon}_{t-h} = \boldsymbol{\mu} + \sum_{h=0}^{\infty} \Theta_h \boldsymbol{\varepsilon}_{t-h}, \quad (26)$$

where

$$\Theta_h := \Psi_h P \in \mathbb{R}^{K \times K}. \quad (27)$$

A.3.4 Impulse response functions

Consider a one-unit shock to variable j at time t , i.e., $\boldsymbol{\varepsilon}_t = \mathbf{e}_j$ (the j -th basis vector) and $\boldsymbol{\varepsilon}_s = \mathbf{0}$ for $s \neq t$. The effect on $\mathbf{y}_{t+h}$ for horizon $h \geq 0$ is

$$\Delta \mathbf{y}_{t+h} = \Theta_h \mathbf{e}_j = (\text{the } j\text{-th column of } \Theta_h). \quad (28)$$

The *impulse response function* from structural shock j to variable k at horizon h is thus

$$\text{IRF}_{k \leftarrow j}(h) = [\Theta_h]_{k,j}. \quad (29)$$

In practice, we compute $\widehat{\Psi}_h$ from the estimated VAR coefficients via eq. (22), form $\widehat{\Theta}_h = \widehat{\Psi}_h \widehat{P}$, and then find $\widehat{\text{IRF}}_{k \leftarrow j}(h)$ for $h = 0, \dots, H$.

A.4 Granger Causality Test Statistic

A.4.1 Definition

Let $y_{k,t}$ and $y_{j,t}$ be two components of the vector $\mathbf{y}_t$. Informally, $y_{j,t}$ *Granger-causes* $y_{k,t}$ if past values of $y_{j,t}$ contain information that helps predict $y_{k,t}$ beyond the information contained in past values of $y_{k,t}$ and other variables.

More precisely, in the context of a VAR(p), we compare two models for the k -th equation:

- **Unrestricted model (U):** includes lags of $y_{j,t}$:

$$y_{k,t} = \beta_0 + \sum_{\ell=1}^K \sum_{i=1}^p \beta_{\ell,i}^{(U)} y_{\ell,t-i} + \boldsymbol{\gamma}^\top \mathbf{d}_t + u_{k,t}^{(U)}. \quad (30)$$

- **Restricted model (R):** excludes all p lags of $y_{j,t}$, i.e., imposes the linear restrictions $\beta_{j,1}^{(U)} =$

$$\dots = \beta_{j,p}^{(U)} = 0:$$

$$y_{k,t} = \beta_0 + \sum_{\ell=1}^K \sum_{i=1}^p \beta_{\ell,i}^{(R)} y_{\ell,t-i} \quad \text{with } \beta_{j,i}^{(R)} = 0 \text{ for all } i + \boldsymbol{\gamma}^\top \mathbf{d}_t + u_{k,t}^{(R)}. \quad (31)$$

The *null hypothesis* of no Granger causality from $y_{j,t}$ to $y_{k,t}$ is

$$H_0 : \beta_{j,1}^{(U)} = \beta_{j,2}^{(U)} = \dots = \beta_{j,p}^{(U)} = 0, \quad (32)$$

which says that lags of $y_{j,t}$ do not enter the k -th equation.

A.4.2 F-statistic for linear restrictions

Let $T^\star = T - p$ denote the effective sample size for the regression. For the k -th equation, denote the OLS residuals from the unrestricted and restricted regressions by $\hat{u}_{k,t}^{(U)}$ and $\hat{u}_{k,t}^{(R)}$, respectively, and define the residual sums of squares

$$\text{RSS}_U = \sum_{t=p+1}^T (\hat{u}_{k,t}^{(U)})^2, \quad \text{RSS}_R = \sum_{t=p+1}^T (\hat{u}_{k,t}^{(R)})^2. \quad (33)$$

Let q be the number of restrictions being tested; here $q = p$ because we restrict p lag coefficients of $y_{j,t}$ in the k -th equation. Let m_U be the number of parameters in the unrestricted equation (including intercept and deterministic terms). The usual F -statistic is

$$F = \frac{(\text{RSS}_R - \text{RSS}_U)/q}{\text{RSS}_U/(T^\star - m_U)}. \quad (34)$$

Under H_0 and standard regularity conditions, F has an F -distribution with $(q, T^\star - m_U)$ degrees of freedom:

$$F \sim F_{q, T^\star - m_U}. \quad (35)$$

We can then obtain a p -value by comparing the observed F to this distribution, and reject the null of no Granger causality if the p -value is less than a chosen significance level (e.g., $\alpha = 0.05$).

In matrix form, multiple Granger tests can be performed simultaneously using the estimated coefficient matrices $\hat{A}_i$ and the estimated innovation covariance $\hat{\Sigma}_u$, but in our implementation we apply eq. (34) equation-by-equation for the sake of interpretability.

A.5 Residual Bootstrap for IRFs

Analytic IRF confidence intervals are derived under assumptions such as correct VAR specification, Gaussian and homoskedastic errors, and large-sample asymptotics. In our application, these assumptions are not fully satisfied. To obtain inference that is robust to such deviations, we use a residual bootstrap, which approximates the sampling variability of the IRFs directly from the

observed data without relying on these parametric assumptions.

A.5.1 Algorithm

Let $\{\mathbf{y}_t\}_{t=1}^T$ be the observed series, and let $\hat{A}_1, \dots, \hat{A}_p, \hat{B}$, and $\hat{\Sigma}_u$ be the parameters estimated by OLS as described in appendix A.1. Let $\hat{\mathbf{u}}_t$ be the corresponding residuals for $t = p + 1, \dots, T$.

Now consider B bootstrap replications. For replication $b = 1, \dots, B$:

- (B1) Resample residuals.** Construct a bootstrap residual sequence $\{\mathbf{u}_t^{*(b)}\}_{t=p+1}^T$ by drawing with replacement from the set of centered residuals $\{\hat{\mathbf{u}}_t - \bar{\mathbf{u}} : t = p + 1, \dots, T\}$, where

$$\bar{\mathbf{u}} = \frac{1}{T^*} \sum_{t=p+1}^T \hat{\mathbf{u}}_t. \quad (36)$$

Centering enforces zero mean in the bootstrap innovations. Because the distribution of the residuals is taken from the data rather than assumed, this step avoids parametric assumptions such as normality.

- (B2) Initialize bootstrap series.** Choose starting values for the bootstrap series $\mathbf{y}_1^{*(b)}, \dots, \mathbf{y}_p^{*(b)}$.
Weset

$$\mathbf{y}_t^{*(b)} = \mathbf{y}_t, \quad t = 1, \dots, p, \quad (37)$$

Using the observed history ensures that each bootstrap sample shares the same starting conditions as the original dataset.

- (B3) Simulate bootstrap series.** For $t = p + 1, \dots, T$, generate

$$\mathbf{y}_t^{*(b)} = \hat{B} \mathbf{d}_t + \sum_{i=1}^p \hat{A}_i \mathbf{y}_{t-i}^{*(b)} + \mathbf{u}_t^{*(b)}. \quad (38)$$

This produces a bootstrap sample $\{\mathbf{y}_t^{*(b)}\}_{t=1}^T$.

- (B4) Re-fit VAR to bootstrap sample.** Estimate a VAR(p) on $\{\mathbf{y}_t^{*(b)}\}$ in the exact same way as for the original data, obtaining $\hat{A}_i^{*(b)}$, $\hat{B}^{*(b)}$ and $\hat{\Sigma}_u^{*(b)}$.

- (B5) Compute bootstrap IRFs.** From $\{\hat{A}_i^{*(b)}\}$ and $\hat{\Sigma}_u^{*(b)}$, compute the bootstrap VMA coefficients $\hat{\Psi}_h^{*(b)}$ for $h = 0, \dots, H$ using eq. (22), obtain a bootstrap Cholesky factor $\hat{P}^{*(b)}$ such that $\hat{\Sigma}_u^{*(b)} = \hat{P}^{*(b)} \hat{P}^{*(b)\top}$, form $\hat{\Theta}_h^{*(b)} = \hat{\Psi}_h^{*(b)} \hat{P}^{*(b)}$, and extract bootstrap IRFs

$$\widehat{\text{IRF}}_{k \leftarrow j}^{*(b)}(h) = [\hat{\Theta}_h^{*(b)}]_{k,j}, \quad h = 0, \dots, H. \quad (39)$$

After B replications, we get distributions for each IRF entry and horizon:

$$\left\{ \widehat{\text{IRF}}_{k \leftarrow j}^{*(b)}(h) \right\}_{b=1}^B. \quad (40)$$

A $(1 - \alpha)$ confidence interval at horizon h can be gotten using quantiles, e.g. the percentile method:

$$\left[q_{\alpha/2}, q_{1-\alpha/2} \right], \quad (41)$$

where q_τ is the τ -quantile of the bootstrap sample. Because these intervals are derived from the empirical distribution rather than asymptotic approximations, they remain valid even when standard VAR assumptions (such as normality or homoskedasticity) are violated.

A.6 Residual Bootstrap for Granger Causality

We also use a residual bootstrap to obtain empirical distributions and p -values for the Granger causality F -statistics in eq. (34). The goal is to simulate many bootstrap series using the fitted VAR model, re-fit the model to each bootstrap series, recompute the F -statistics, and compare the observed statistic to this bootstrap distribution.

Let F_{obs} be the observed F -statistic for a given Granger test (e.g., from variable j to k). Using the same bootstrap series $\{\mathbf{y}_t^{*(b)}\}_{t=1}^T$ generated in appendix A.5, we do:

- (G1) **Re-fit VAR to bootstrap series.** For replication b , estimate the VAR(p) on $\{\mathbf{y}_t^{*(b)}\}$, obtaining $\{\hat{A}_i^{*(b)}\}$ and $\hat{B}^{*(b)}$.
- (G2) **Compute bootstrap F -statistic.** For the Granger relationship of interest (e.g., $j \rightarrow k$), re-estimate the unrestricted and restricted versions of the k -th equation on the bootstrap series and compute $F^{*(b)}$ using the same formula as in eq. (34).

This yields bootstrap statistics

$$\{F^{*(b)}\}_{b=1}^B. \quad (42)$$

A *bootstrap p -value* for the observed F_{obs} is then

$$\hat{p}_{\text{boot}} = \frac{1 + \#\{b : F^{*(b)} \geq F_{\text{obs}}\}}{B + 1}. \quad (43)$$

If $\hat{p}_{\text{boot}} < \alpha$ (e.g., $\alpha = 0.05$), we regard the Granger relationship as statistically significant at level α , based on the bootstrap approximation to the finite-sample distribution.

The bootstrap framework in appendices A.5 and A.6 thus provides a unified way to quantify uncertainty in both impulse responses and Granger causality tests, using the same core VAR model and residual structure.

B Additional Figures and Tables

C Author Contributions